Cross-Domain Self-Training Network With Prototype Rectification for Few-Shot Hyperspectral Image Classification

Qingyan Wang, *Member, IEEE*, Peng Luo, Junping Zhang, *Senior Member, IEEE*, Zhenhang Yao, and Yujing Wang

***Abstract*—In recent years, the classification research of hyperspectral image (HSI) with scarce labeled samples has made significant progress, demonstrating strong performance when labeled samples are limited. However, current methods often overlook the importance of class prototypes. In addition, while conditional domain adversarial adaptation (CDAA) is employed to mitigate domain shift, it fails to account for the varying degrees of data distribution discrepancies across different classes, leading to suboptimal alignment. To address these issues, a cross-domain self-training network with prototype rectification (CSNPR) for few-shot HSI classification is proposed. First, a 3-D ghost attention network with Brownian distance covariance (TGAN-B) is introduced to improve the representation learning of HSI, generating more effective class prototypes. Next, an adaptive weighted CDAA (ACDAA) strategy is proposed, which assigns weights based on the domain discrepancy for better alignment. Finally, a pseudo label selection (PLS) module is developed to select the most confident pseudo-labeled samples from each class in the target domain (TD), thereby enriching the training set. The experimental findings derived from the analysis of three publicly available HSI datasets prove that the CSNPR method outperforms existing state-of-the-art (SOTA) methods, achieving superior and stable performance even with sparse labeled samples. Our source code is available at https://github.com/Yzh190-hue/CSNPR**

***Index Terms*—Domain adaptation (DA), few-shot learning (FSL), hyperspectral image classification (HSIC), self-training Brownian distance covariance (BDC).**

I. Introduction

HYPERSPECTRAL image (HSI) consists of hundreds of continuous spectral bands captured over the same region, combining both spectral and spatial information [1], [2]. Compared to traditional color images, HSI offers rich geospatial and spectral details, enabling more precise identification of ground objects. This capability has significant applications in areas such as land cover classification [3], resource surveying [4], and ecological monitoring [5]. HSI classification (HSIC) remains a key application in remote sensing and continues to be a prominent research focus in optics.

Early HSIC methods are typically divided into two primary parts: feature extraction and classification. Traditional feature extraction techniques, including principal component analysis [6], linear discriminant analysis [7], and manifold learning [8], aim to extract essential information from raw HSI. For classification methods, such as k-nearest neighbors [9], support vector machines [10], and random forests [11] are commonly employed. However, these approaches focus on individual pixels while overlooking the correlations that exist between neighboring pixels.

With the advent of deep learning, HSIC has made significant progress, and the convolutional neural network (CNN) has been particularly extensively applied in these tasks. First, Hu et al. [12] applied 1-D convolution kernels to process spectral information. Then, Makantasis et al. [13] proposed a 2-D CNN to extract spatial features from HSI. Later studies also demonstrated the importance of incorporating spatial information for enhanced classification performance [14]. Li et al. [15] further advanced the field with a 3-D CNN, capable of simultaneously extracting spectral and spatial information. These methods underscore the potential of deep learning in HSIC. In addition, models incorporating advanced network structures, such as residual network [16], dense connection [17], and attention mechanism [18], have superior classification performance. While deep learning methods have shown great performance due to their powerful feature extraction ability, they generally require substantial amounts of labeled samples for training. Nevertheless, the scarcity of labeled samples, driven by the complexity of HSIs imaging and the challenge of manual annotation, considerably curtails the practical implementation of these methodologies.

Transfer learning is a common approach to addressing the scarcity of labeled samples. Domain adaptation (DA) methods aim to leverage abundant labeled data from the source domain (SD) to improve classification performance in a target domain (TD) with limited annotations. Long et al. [19]. were the first to introduce deep neural networks (DNNs) into DA, proposing a framework that employs three adaptive layers based on multikernel variants of the maximum mean discrepancy to

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align the edge feature distributions across domains. Wang et al. [20]. introduced the domain adversarial broad adaptation network (DABAN) for HSIC, which effectively aligns the statistical distributions between the SD and TD while capturing more representative domain-invariant features. Yu et al. [21]. further enhanced DA by proposing a content alignment approach within an adversarial learning framework, aiming to achieve finer semantic consistency between domains. However, most DA approaches assume that the classes in two domains are consistent. Once a new class appears in the TD, the model's performance will degrade significantly.

The second approach to addressing the scarcity of labeled samples is few-shot learning (FSL), which employs a meta-learning strategy to train models and divides the dataset into different meta tasks to learn common knowledge. Importantly, it does not require different domains to share the same categories. Recent research in the field of FSL has concentrated on metric-based FSL approaches for HSIC, which achieve classification by measuring the distance between the support set and the query set. Liu et al. [22]. proposed a deep FSL (DFSL) method to accurately achieve classification with sparse labeled samples. However, this method has no consideration of the problem of domain shift. To tackle this issue, Li et al. [23] advanced a deep cross-domain FSL (DCFSL) method, integrating the DA approach based on adversarial learning with the FSL method, reducing domain shift within the same framework. Despite this, DCFSL relies on local spatial information for domain alignment, overlooking nonlocal spatial information with strong correspondence. To overcome this limitation, Zhang et al. [24] proposed the graph information aggregation cross-domain FSL (Gia-CFSL) approach. It combines graph-based information aggregation for cross-domain FSL and domain alignment to address the aforementioned shortcoming. The FSL methods described above typically use Euclidean distance to measure the distance between features in the embedding space. However, the Euclidean distance metric assumes that feature dimensions are independent of variance, and it is less sensitive to the distribution of intra-class sample features relative to prototype sample features. Therefore, Xi et al. [25] proposed an FSL method with a class-covariance metric (CMFSL), which employs the Mahalanobis distance metric to replace Euclidean distance and learn domain-invariant class representations without additional classifiers. As highlighted by Lei et al. [26], recently, FSL methodologies frequently fail to accurately represent prototypes of all classes. To address this shortcoming, the dual-adjustment mode-based cross-domain meta-learning (DMCM) approach has been proposed. This method aims to generate more precise class prototypes and mitigate the domain shift.

The quality of the embedding space learned by the metric-based FSL method depends on the feature extractor. However, the existing approaches for quantifying the dependencies between hyperspectral embedding features either model only the marginal distribution while neglecting the joint distribution, thereby limiting the expressiveness of the feature representation, or consider the joint distribution at higher computational costs. In addition, the FSL methods discussed above employ a conditional domain adversarial adaptation (CDAA) strategy to reduce domain shift. However, when performing domain alignment, the varying distributional differences across classes between the two domains are often overlooked. The global alignment scheme is suboptimal, as it may negatively affect the already aligned features.

The third approach to addressing the scarcity of labeled samples is semi-supervised learning. Wu and Prasad [27] focused on assigning pseudo labels to unlabeled data, further reducing the reliance on labeled samples. Zhang et al. [28] proposed a cross-domain self-taught network (CDSTN), which integrates DA with a semi-supervised self-teaching strategy to achieve FSL. In this method, a trained network is employed to assign pseudo-labels to unlabeled samples, and a module for pseudo-label refinement is introduced to select pseudo-labeled samples with high confidence for each class, thereby enriching the training set in TD. The selection of pseudo-labeled samples significantly impacts the performance of the module. Consequently, identifying pseudo-labeled samples accurately becomes a key challenge in a semi-supervised strategy. Most existing methods focus on leveraging knowledge of SD to enhance the accuracy of pseudo labels assigned to TD, while often neglecting the internal relationships among samples of the same class within the TD.

To address the scarcity of labeled samples and challenges discussed above, this article proposes a cross-domain self-training network with prototype rectification (CSNPR) for few-shot HSIC. The CSNPR method combines metric-based FSL, the CDAA strategy, and a semi-supervised strategy, and the following specific contributions are hereby enumerated.

1) Building upon the DMCM, Brownian distance covariance (BDC) is introduced as a low-cost yet comprehensive method to quantify the dependency between features, realizing greater representation of HSI and generating more accurate class prototypes.
2) Adaptive weights are incorporated into the traditional CDAA strategy, allowing the model to pay more attention to classes exhibiting great data distribution differences between SD and TD.
3) A pseudo label selection (PLS) module is constructed to identify the pseudo-labeled samples with a high degree of confidence and greater representativeness, thereby improving the quality of training samples.

The remainder of this article is organized as follows. Section II presents the principle of CSNPR. Section III outlines the experimental setup and analyzes results. Section IV concludes the study.

II. Methodology

A. Overall Framework

As illustrated in Fig. 1, the proposed model consists of four main components: a feature extraction module, an FSL module, a domain alignment module, and a PLS module. During training, two types of FSL are employed, one tailored for the SD and the other for the TD. These FSL processes are alternated iteratively, enabling the model to simultaneously acquire generalizable knowledge from the SD and learn a discriminative embedding space that effectively determines whether input samples belong to the TD classes.

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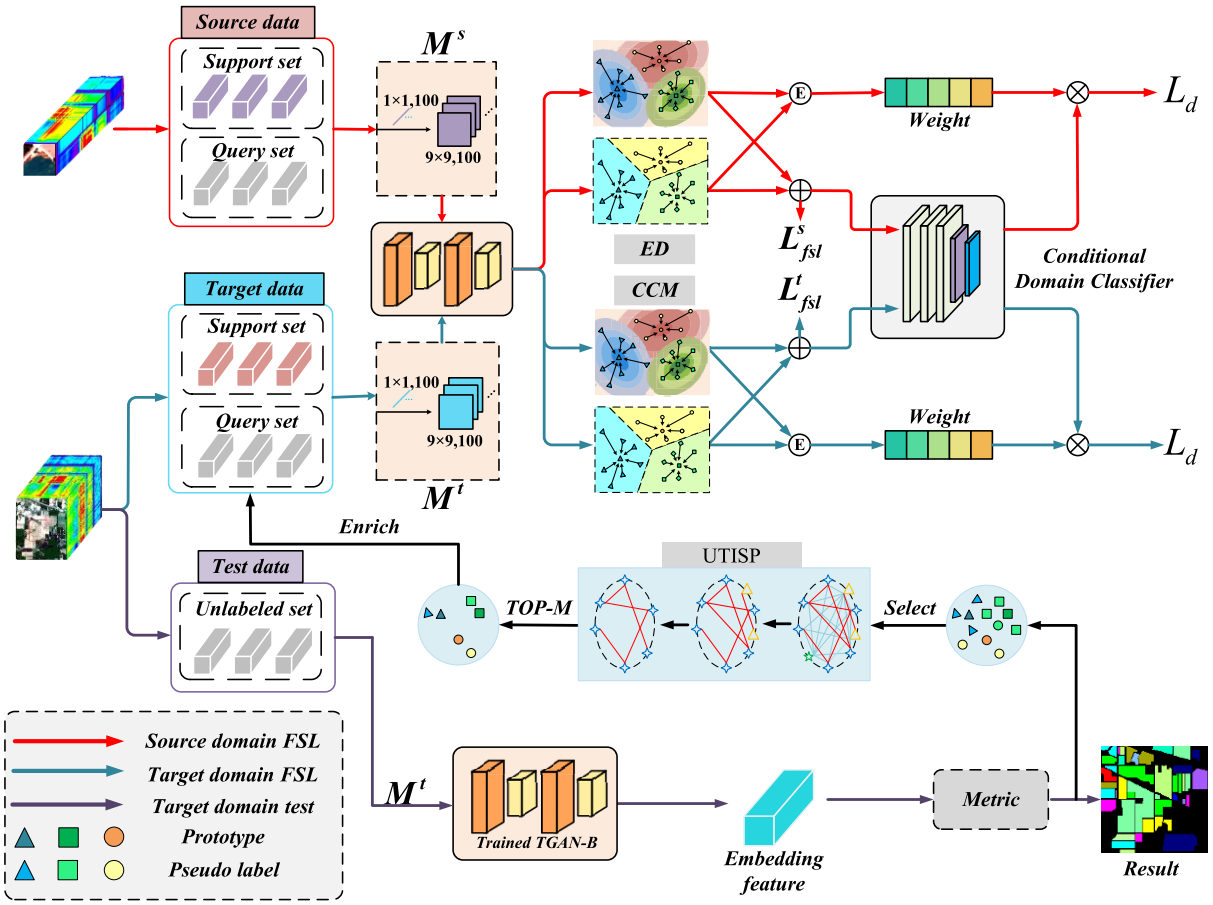


Fig. 1. CSNPR framework.

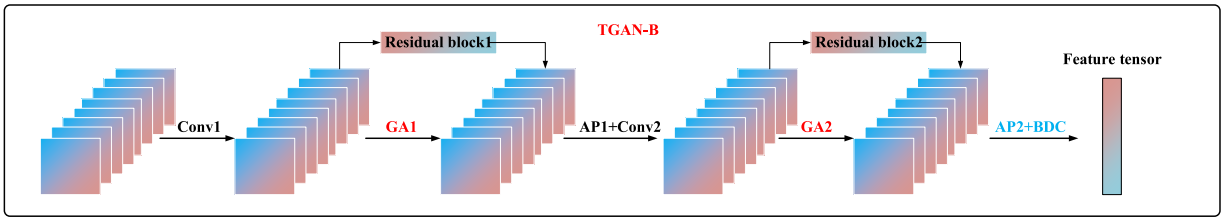


Fig. 2. Structure of the TGAN-B.

The testing process of unlabeled samples, depicted at the bottom of Fig. 1, proceeds as follows. First, the trained feature extractor is utilized to extract spatial-spectral features from both labeled and unlabeled samples. The prototype features of each class are then generated from the features of labeled samples. Subsequently, the metric module measures the distances between the features of unlabeled samples and the prototype features of all classes. Then, pseudo labels are assigned according to these distances, providing an initial classification result. Finally, pseudo-labeled samples are processed through the PLS module, where high-confidence pseudo-labeled samples of each class are selected. And these samples will enrich the labeled dataset of the TD and take part in the next training process.

B. Feature Extraction

The feature extraction component primarily consists of two mapping layers and an embedded feature extractor. HSIs captured by different sensors often have different spectral dimensions. To ensure that data from two domains can be processed by the same embedded feature extractor, two mapping layers are employed to standardize the spectral dimensions of HSI data across two domains. The mapping layers are composed of a simple 2-D CNN.

The embedded feature extractor section incorporates a BDC module on top of the 3-D ghost attention (GA) network, and the structure of the 3-D GA network with BDC (TGAN-B) is shown in Fig. 2. The core of the TGAN-B is the GA module. As illustrated in Fig. 3, it comprises the Ghost module [29] and the normalization-based attention module (NAM) [30]. The Ghost module is designed to replace some convolution operations with cheap operations, thereby generating numerous ghost feature maps at a lower computational cost. Meanwhile, NAM can reassign feature weights to emphasize key features, further enhancing the extraction of discriminative features.

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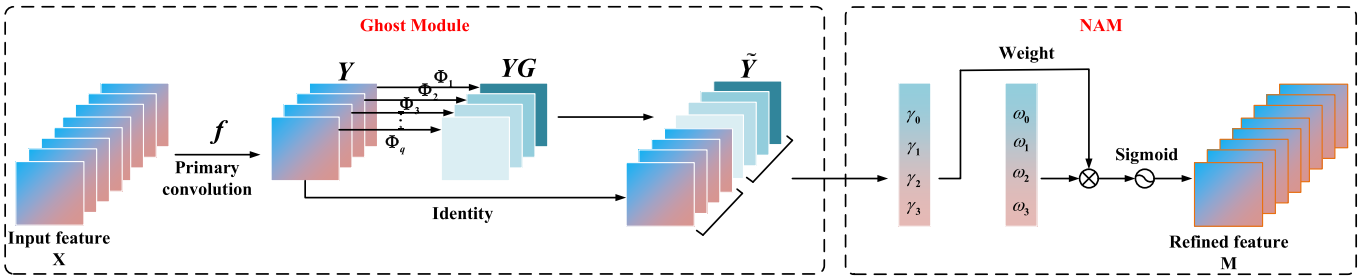


Fig. 3. Structure of the GA module.

Domain discrepancies are often reflected in higher order dependencies among spectral bands. BDC measures the statistical dependence between two random variables by characterizing their joint characteristic functions and equals zero if and only if the variables are independent.

By incorporating BDC, the proposed method can effectively model complex inter-feature dependencies, thereby enhancing distribution-level alignment beyond simple mean-based constraints. BDC is defined as the Euclidean distance between the joint characteristic function of two random vectors and the product of their marginal distributions [31]. It can quantify the dependency between two random variables. Given $X \in R^n$ and $Y \in R^n$, which are variables of dimension m and n, respectively. Their BDC measure is denoted as $\rho(X, Y)$. P is the BDC matrix of variable X, with its element p_{kl}. Q is the BDC matrix of variable Y, with its element q_{kl}.

The BDC module is designed based on BDC theory and consists of two layers: one performs dimensionality reduction, and the other computes the BDC matrix. Notably, the length of the output vector from the BDC module increases quadratically with the number of input channels. To prevent the output vector from becoming excessively large, a convolutional layer is applied for dimensionality reduction before the computation of the BDC matrix.

For the computation, BDCPool layer reshapes the dimension of the embedding feature $V \in R^{c\times h\times w}$ into $V \in R^{c\times hw}$ first, and each column $v \in R^{hw}$ can be regarded as a discrete observation of V. This process can be applied to the BDC theory for calculation. The following formula is employed to calculate $\rho(X, Y)$:

$$\tilde{P} = 2\left(1\left(V^T V \circ I\right)_{sym} - 2V^T V\right) \tag{1}$$

$$\widehat{P} = \left(\sqrt{\tilde{p}_{kl}}\right) = \left(\sqrt{\|v_k - v_l\|^2}\right) \tag{2}$$

$$P = \widehat{P} - \frac{2}{c}\left(1\widehat{P}\right)_{sym} + \frac{1}{d^2}1\widehat{P}1 \tag{3}$$

$$\rho(X, Y) =< p, q >= p^T q \tag{4}$$

where $\tilde{p}_{kl}$ is the element of matrix $\tilde{P}$. $\circ$ represents the Hadamard product operation of matrices. $\|\cdot\|$ represents the Euclidean norm. d represents the reduction dimension, which also corresponds to the row (or column) of the BDC matrix. and matrix $1 \in R^{c\times c}$ is a matrix with all elements being one. $<\cdot>$ represents the matrix inner product and matrix I is the identity matrix. $(Matrix)_{sym} = 1/2(Matrix + Matrix^T)$.

Then, upper triangular extraction and vectorization operations are performed after the computation of the BDC matrix to obtain the final vector with the length of $d(d+1)/2$. Table I shows the detailed configuration of the TGAN-B network structure.

TABLE I
CONFIGURATIONS OF THE TGAN-B

Layer	Operation	Kernel	Output Size
Input	*N/A*	*N/A*	(9×9×B,1)
DR	Conv2D	(1,1)	(9×9×100,1)
Conv1	Conv3D	(3,3,3)	(9×9×100,N)
GA1	Conv3D	(1,1,1)	(9×9×100, 2N/r)
	Conv3D	(3,3,3)	(9×9×100, 2N×(r-1)/r)
	Concatenate	*N/A*	(9×9×100, 2N)
	NAM	*N/A*	(9×9×100, 2N)
Residual Block1	Conv3D	(1,1,1)	(9×9×100, 2N)
	Add	*N/A*	*N/A*
AP1+Conv2	AvgPool3D	(4,1,1)	(9×9×25, 2N)
	Conv3D	(1,1,1)	(9×9×25, 2N)
GA2	Conv3D	(1,1,1)	(9×9×25, 4N/r)
	Conv3D	(3,3,3)	(9×9×25,4N×(r-1)/r)
	Concatenate	*N/A*	(9×9×25, 4N)
	NAM	*N/A*	(9×9×25, 4N)
Residual Block2	Conv3D	(1,1,1)	(9×9×25, 4N)
	Add	*N/A*	*N/A*
AP2	AvgPool3D	(4,1,1)	(9×9×6, 4N)
BDCPool	Conv2D	(1,1)	(9×9×d,1)
	BDC	*N/A*	(d(d+1)/2,1)

Compared with CNN, TGAN-B can more effectively model long-range dependencies between spectral bands and spatial regions. Moreover, by incorporating the Ghost module, TGAN-B significantly reduces redundant feature computation while maintaining strong representation capability, achieving a favorable balance between feature expressiveness and computational efficiency. Compared with transformer-based networks, TGAN-B avoids the high computational cost associated with self-attention over large token sequences and can achieve stable performance without relying on large-scale training data.

C. FSL Module

The approach used in this study is a metric-based FSL method. In each iteration, the samples generate feature vectors by passing through the mapping layer and the embedded feature extractor. The features of the support set are then used to generate the prototype features for each class. Subsequently,

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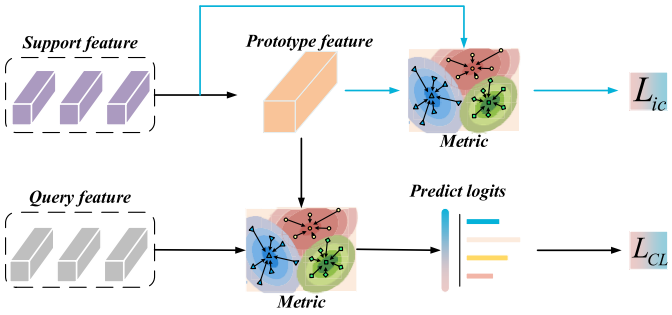


Fig. 4. Schematic of new FSL strategy.

the distance between the features of the query set and the prototype features is computed in the metric space to predict the classification results. Finally, the network parameters are iteratively updated such that the mapped features, after passing through the embedded feature extractor, achieve the following objective: within the same metric space, the distance between the features in the query set and the prototype features of the same class is minimized, while the distance between the features of query samples and all prototype features is maximized. The predicted results, after processing through the metric space, can be represented by the following equation:

$$P\left(\hat{y}_i | x_i \in Q_s\right) = SoftMax\left(-M\left(f\left(x_i\right), c^k\right)\right) \tag{5}$$

where x_i is the sample from the query set, and $\hat{y}_i$ is the predicted label of x_i. $SoftMax(\cdot)$ represents the SoftMax function. $f(\cdot)$ is the mapping function of embedded feature extractor, and c^k is the prototype of the kth class of samples. $M(\cdot)$ can be any suitable distance metric, such as cosine distance metric, euclidean distance metric, and class covariance metric (CCM).

Take the FSL of SD as an example. After calculating the probability of the query set, the loss of the SD is computed through the cross-entropy loss, as shown in the following formula:

$$L_{fsl}^s = L_{CL}^s = E_{S_s, Q_s}\left[-\sum_{(x,y)\in Q_s} \log P_\varphi^s\left(y_{js} = k | x_{js}\right)\right] \tag{6}$$

where y_{js} represents the predicted label of the jth sample x_{js}, which belongs to SD. $\varphi(\cdot)$ denotes the feature extractor, P_φ^s represents the predicted class probability in the SD based on the extracted feature $\varphi(\cdot)$.

The aforementioned FSL strategy does not pay enough attention to the class prototype. To obtain a more effective class prototype, this study introduces a novel FSL strategy that incorporates the prediction loss of support samples into the loss function, thereby increasing the mapping distances between class prototypes in the metric space. The final strategy is illustrated in Fig. 4.

The new loss function of SD is as follows:

$$L_{fsl}^s = L_{CL}^s + L_{ic}^s \tag{7}$$

$$L_{ic}^s = E_{S_s}\left[-\sum_{(x,y)\in S_s} \log P_\varphi^s\left(y_{is} = k | x_{is}\right)\right]. \tag{8}$$

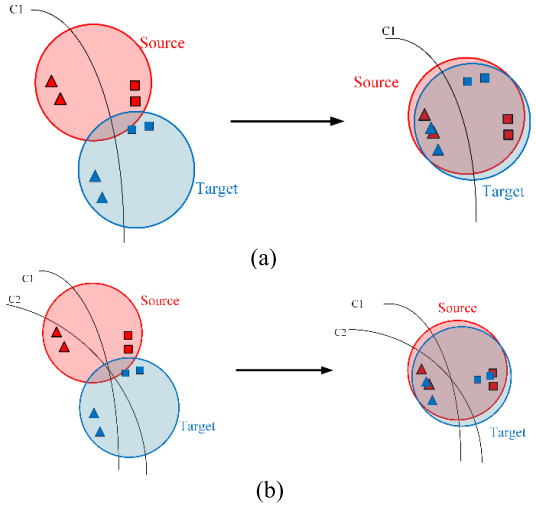


Fig. 5. Schematics of two CDAA strategies. (a) Schematic of the new CDAA strategy. (b) Schematic of the improved CDAA strategy.

D. Adaptive CDAA Strategy

In conditional adversarial DA strategies, the multimodal structure can be jointly modeled by combining the feature extractor f and the classifier predictions g. Consequently, the conditional adversarial DA network is capable of capturing the multimodal structure underlying complex data distributions. The proposed network can be framed as a minimax optimization problem involving two competing objectives: 1) minimize $E(F, G)$ with respect to the classifier G, to reduce the classification risk on the SD and 2) minimize $E(D, F, G)$ with respect to the domain discriminator D, aiming to accurately distinguish between the SD and TD; meanwhile, maximize $E(D, F, G)$ with respect to the feature extractor F and classifier G, to encourage domain-invariant feature learning. The computation of $E(D, F, G)$ is as follows:

$$\min_D \max_{F,G} E\left(D, F, G\right) = -E_{x_i^s \sim P_s(x)} \log\left[D\left(f_i^s, g_i^s\right)\right] - E_{x_j^t \sim P_t(x)} \log\left[1 - D\left(f_j^t, g_j^t\right)\right]. \tag{9}$$

As illustrated in Fig. 5(a), when notable disparities exist in the class-wise data distributions between the SD and TD, for example, class A may exhibit only minor shifts, whereas class B presents substantial divergence. The traditional conditional adversarial domain alignment strategy reduces the distributional gap for class B, but it may inadvertently widen the gap for class A. This outcome occurs because the conventional strategy aligns domains from an overall perspective without taking into account the varying distributional differences among classes.

Inspired by Luo et al. [32], we add an adaptive weight to the traditional CDAA strategy, encouraging the model to focus on classes with significant differences between the SD and TD. However, it is insufficient to achieve that by a single metric space. Therefore, this section employs both Euclidean distance and CCM when learning the metric space. Each metric produces its own metric space and corresponding prediction outcomes, both of which can be adjusted based on spatial-spectral features. Specifically, the two metric spaces assess the features generated by F, and a classifier subsequently produces

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two class distributions, P_{ED} and P_{CCM}. Then, they are fused to obtain the final class, the distribution P. The new strategy is shown in Fig. 5(b)

$$P(\hat{y}_i|x_i) = SI^* P(\hat{y}_i|x_i)_{ED} + (1 - SI)^* P(\hat{y}_i|x_i)_{CCM} \tag{10}$$

where SI is the similarity between P_{ED} and P_{CCM}. The computation of SI is as follows:

$$SI(P_{ED}, P_{CCM}) = \frac{ED(P_{ED}, P_{CCM})}{\sqrt{2}}. \tag{11}$$

The new conditional adversarial domain loss can be expressed as follows:

$$\begin{aligned}\min_D \max_{F,G} L_{DA} = &-E_{x_i^s \sim P_s(x)} \log\left[D\left(f_i^s, g_i^s\right)\right] \\ &- E_{x_j^t \sim P_t(x)}\left[(\lambda_1 SI + \xi) \log\left[1 - D\left(f_j^t, g_j^t\right)\right]\right.\end{aligned} \tag{12}$$

where λ_1 controls the weight of adaptive adversarial loss and ξ controls the stability of the training process. During the conditional alignment process, to address the class imbalance caused by varying sample sizes, class-wise weights are introduced to emphasize the importance of minority classes. The weight can be formulated as follows:

$$\omega = \frac{1}{\log(n_i + 1)} \tag{13}$$

where n_i denotes the number of samples in the i-th class. Classes with a larger number of samples are assigned smaller weights, while minority classes are given larger weights to alleviate the distortion in domain alignment caused by class imbalance. Finally, the calculation of the entire SD loss is as follows:

$$L_{total}^s = L_{fsl}^s + \lambda_2 L_{DA} \tag{14}$$

where λ_2 controls the relative proportion of adversarial loss in the two domains to the overall loss function. The loss computation in the TD follows a similar formulation. The hyperparameters $\lambda_1\lambda_2$ and ξ are determined without using target-domain labels. Specifically, they are selected based on source-domain validation and then fixed across all TDs to evaluate the generalization ability of the proposed method.

Equations (11) and (12) reveal that when SI is high, the corresponding Euclidean distance is large. In this scenario, the spatial-spectral features of samples within the same class exhibit dissimilar class distributions across different metric spaces and do not share similar joint distributions between the SD and TD. Consequently, the loss associated with that class occupies a larger proportion of the overall conditional adversarial domain loss, prompting the model to focus more on aligning that class. Conversely, when SI is low, the Euclidean distance is small, and the spatial-spectral features of samples within the same class display similar class distributions. In this case, the corresponding loss contributes less to the overall conditional adversarial domain loss, and the model does not allocate excessive attention to that class. Thus, the proposed adaptive weighted CDAA (ACDAA) strategy is better at aligning classes with more pronounced differences between the two domains.

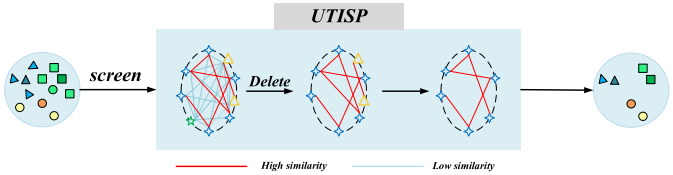


Fig. 6. Flowchart of PLS module.

E. Pseudo Label Selection

Existing pseudo-labeling approaches primarily focus on leveraging knowledge of SD to enhance the accuracy of pseudo labels, often neglecting the intrinsic relationships among samples of the same target class. To address this issue, this section constructs a PLS module that selects high-confidence pseudo-labeled samples based on the similarity among TD samples.

As illustrated in Fig. 6, the PLS module comprises two steps: screening step and using target intra-class similarity to pick (UTISP) step. The screening step is conducted as follows. First, samples are partitioned according to their assigned pseudo labels, with subsequent operations performed on a per-class basis. Subsequently, samples are filtered using thresholds derived from kurtosis and the minimum classification probability, ensuring that the resulting probability distribution more closely approximates a normal distribution rather than a uniform one. The kurtosis is defined as follows:

$$Kurtpsis = E\left[\left(\frac{p - \mu}{\sigma}\right)^4\right] \tag{15}$$

where p is the class-distribution probability of the sample, μ and σ are the mean and variance, respectively.

Finally, samples are ranked in descending order based on their highest predicted probability, with higher ranked samples exhibiting greater confidence. To accommodate class imbalance, the proportion of selected samples is adaptively adjusted based on the number of samples in each class, ensuring representativeness while keeping the computational overhead manageable. After the screening step, the UTISP operation is performed, beginning with the computation of intra-class similarity among samples

$$S_{i,j}^k = \begin{cases} 0, & i = j \\ \dfrac{< Z_i^{T_k}, Z_j^{T_k} >}{\|Z_i^{T_k}\| \|Z_j^{T_k}\|}, & i \neq j \end{cases} \tag{16}$$

where $S_{i,j}^k$ represents the cosine similarity between the ith sample and the jth sample in the kth pseudo class, $i, j \in \{1, 2, \ldots, n_t^k\}$. n_t^k represents the number of labeled samples. $Z_i^{T_k}$ is the ith pseudo-labeled sample of the pseudo class k in the TD. After calculating the similarity, the pseudo-labeled samples with low similarity are removed according to the threshold δ. The calculation of the threshold is as follows:

$$\delta = \eta^* S_{\max}. \tag{17}$$

The hyperparameter η is a coefficient in the PLS module that controls the intra-class similarity threshold. It jointly determines the final selection threshold δ with the intra-class maximum similarity Smax. The value range of η is [0.1, 0.5].

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The calculation method of Smax remains unchanged, and the OA values of the models under different η are tested. When $\eta < 0.3$, δ is too small, resulting in the selection of a large number of low-confidence pseudo-labels; when $\eta > 0.3$, δ is too large, leading to a reduction in the effective training sample size; therefore, when $\eta = 0.3$, it can both select high-confidence samples through $\delta = \eta * S_{\max}$ and retain sufficient training data, balancing the model performance and training stability. Regarding the pseudo-labeled sample Z^{T_k} in the TD as a node, and taking M^k as the adjacency matrix of the undirected graph formed by each node in the ith class, if the similarity between nodes is higher than the threshold, they are neighbors, and the mask label is one. Otherwise, the mask label is zero, as shown in (17)

$$M_{i,j}^k = \begin{cases} 1, & S_{i,j}^k \geq \delta \\ 0, & S_{i,j}^k < \delta. \end{cases} \tag{18}$$

After making the mask label, only the nodes labeled as 0 need to be removed from the undirected graph. However, the pseudo-labeled samples assigned to the kth class may be contaminated by instances from other visually or semantically similar classes. If we select high-confidence pseudo labels solely based on the threshold δ, we might inadvertently include samples from these similar classes as high-confidence samples. Therefore, some modifications are required. Specifically, the adjacency matrix is summed row by row, and the subscript of the maximum element is taken

$$D_i^k = \sum_j M_{i,j}^k \tag{19}$$

$$m = \arg\max_i D_i^k \tag{20}$$

where D_i^k represents the number of nodes adjacent to the ith sample in the kth class. The higher this value, the deeper the internal connection of this sample with samples of the same class. The pseudo-labeled sample $Z_m^{T_k}$ corresponding to the maximum D_i^k value has the most adjacent nodes, showing a high similarity to many samples of the same class. It can represent the pseudo-labeled sample with the highest confidence in the kth class of the TD and is also more representative. Finally, M samples are selected from the nodes adjacent to $Z_m^{T_k}$ as the TOP M samples. These samples are added to the labeled sample set of the TD to expand the training sample set for subsequent training.

III. Experimental Results and Analysis

A. Introduction of Experimental Datasets

The method proposed in this article is a cross-domain FSL model, which learns prior knowledge from the SD and applies it to the TD. The SD requires a dataset with a wide variety of classes and strong diversity, so we select the Chikusei (CH) dataset as the SD. Then, three public hyperspectral remote sensing datasets are chosen as the TD datasets.

1) SD: The aerial hyperspectral dataset employed in this study was acquired using the Headwall Hyperspec-VNIR-C sensor over CH, Japan. It comprises 19 land-cover classes captured across 128 spectral bands, spanning a wavelength range of 343–1018 nm. The image has a spatial resolution of 2.5 m per pixel and a size of 2517 × 2335 pixels.
2) TD: The Indian Pines (IP) dataset was collected by the AVIRIS sensor, originally comprising 224 spectral bands within the 0.4–2.5 μm range. After removing bands affected by water absorption, 200 bands were retained. The image has a spatial resolution of 20 m per pixel and a size of 145 × 145 pixels.

The Pavia University (UP) dataset was acquired using the ROSIS airborne sensor. It initially contained 115 spectral bands, which were reduced to 103 after preprocessing. The image size is 610 × 340 pixels, with a spatial resolution of 1.3 m per pixel.

The WHU-Hi-LongKou (LK) dataset was collected in LK Town, Hubei, China, using a Headwall Nano-Hyperspectral imaging sensor (8 mm focal length) mounted on a DJI M600 Pro UAV platform [33], [34]. The image size is 550 × 400 pixels, and the spatial resolution is approximately 0.463 m per pixel. This dataset includes 270 spectral bands covering the 400–1000 nm wavelength range. To address the large number of samples in the original dataset, 15% of the samples in each class were randomly selected for use in the subsequent experiments.

Fig. 7 presents land cover categories, the corresponding number of samples, maps of ground truth, and pseudo color maps for the IP, UP, and LK datasets.

B. Experiment Setting

All experiments are carried out under the following conditions: Windows11 operating system, with a 13th Gen Intel[1] Core[2] i7-13700H CPU at 2.4 GHz, 16 GB RAM (expandable to 64 GB), and an Nvidia GeForce RTX 4060 Laptop GPU. The Python environment included Python 3.7, PyTorch 1.12.0, and CUDA 11.6.2.

The learning rate is set to 0.001, and the cross-entropy loss function is adopted to measure the classification error. The Adam optimizer is employed to facilitate efficient model training. Xavier initialization is applied to both the convolutional kernels and the linear layers. For the TGAN-B model, the hyperparameters N and r are set to 8 and 4, respectively, while M is fixed at 1. To ensure sufficient convergence, the number of training iterations is set to 10 000. In constructing the SD training dataset, 200 labeled samples are randomly selected from each class. Classes with fewer than 200 labeled samples are excluded. For the TD, L labeled samples per class are randomly chosen for FSL, where $L \in \{1, 2, 3, 4, 5\}$. Each class in the TD follows the same sampling strategy.

Three parameters $\{\lambda_1, \lambda_2, \xi\}$ are set to {10, 1, 0.4}, {0.01, 0.1, 0}, and {0.01, 0.1, 0} in the IP, UP, and LK datasets, respectively. The patch size is set to 9 9. And the dimension of the mapping layer is set to 100. The overall accuracy (OA), average accuracy (AA), and Kappa coefficient based on the confusion matrix are used as experimental indicators. In order to compare more intuitively, the Kappa coefficient results of the experimental results in this article have been magnified 100×. Then, ten experiments were carried out in each method

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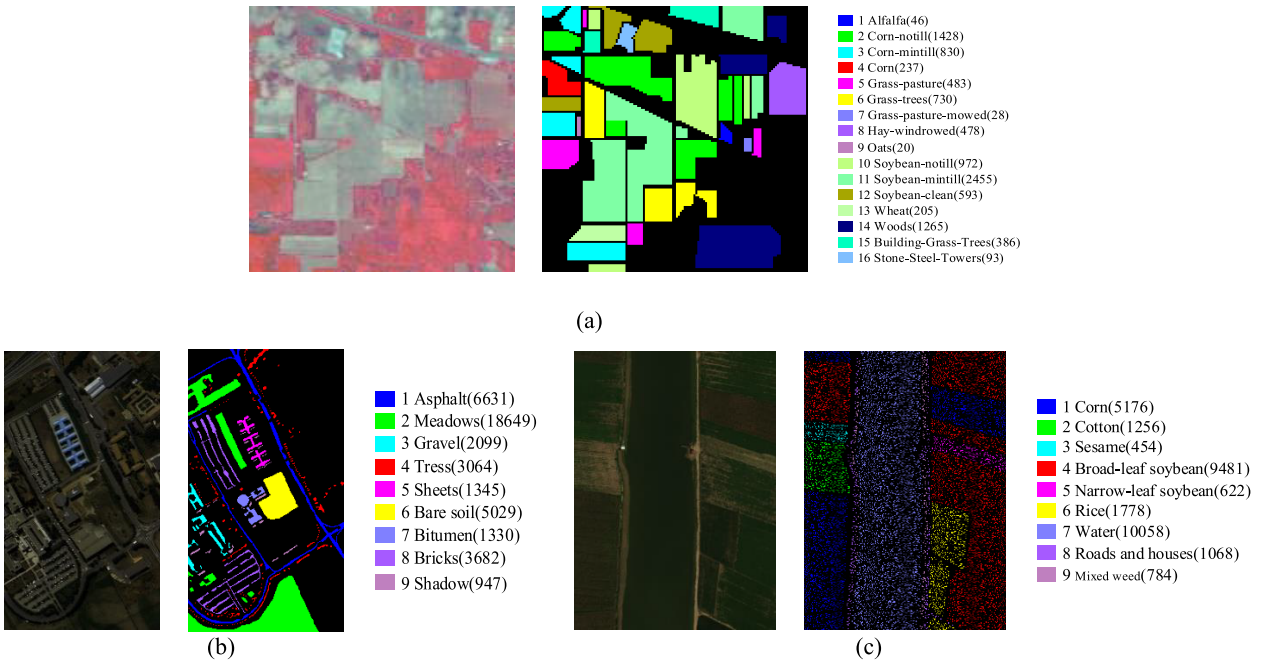


Fig. 7. Truth map and false color map of TD dataset. (a) IP dataset. (b) UP dataset. (c) LK dataset.

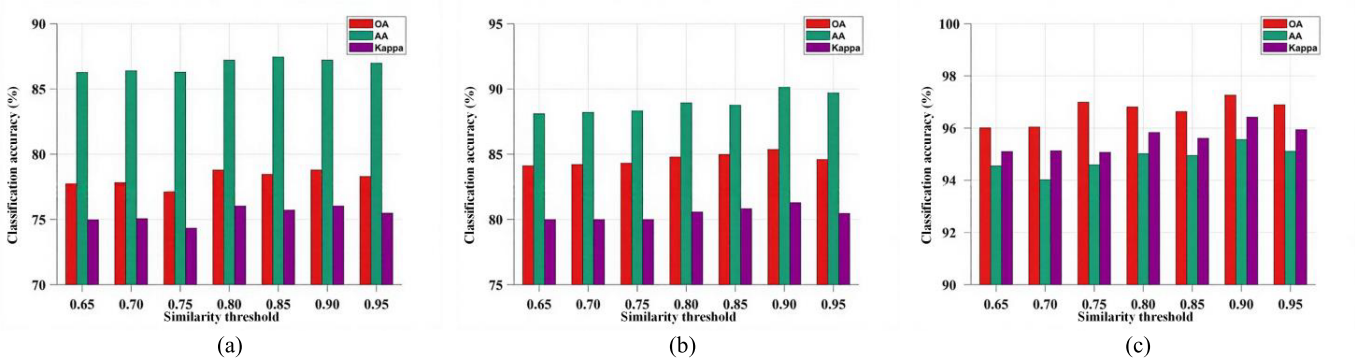


Fig. 8. Classification results of different similarity threshold. (a) IP dataset. (b) UP dataset. (c) LK dataset.

TABLE II
DETERMINATION OF THE PARAMETER d

		$d = 32$	$d = 64$	$d = 128$
IP	OA	76.64	78.79	79.27
	Time(s)	2137.82	3067.49	4217.58
UP	OA	83.07	85.36	85.67
	Time(s)	1862.51	2537.92	3496.47
LK	OA	93.73	97.26	97.69
	Time(s)	2466.73	3508.14	4972.75

to avoid contingency, and different random seeds were selected in each experiment.

In the BDC matrix, the selection of the parameter d is critical, as it requires a balance between feature representation capability and computational complexity. We evaluate three values of d, namely 32, 64, and 128, as shown in Table II. When $d = 32$, the limited capacity to model feature dependencies results in a decrease of approximately 2.1%–3.5% in OA compared to $d = 64$. When $d = 128$, although the OA improves marginally by only 0.3%–0.5%, the computational cost increases by approximately three times, and the training time is extended by about 40%, leading to low cost-effectiveness. Therefore, considering both feature representation effectiveness and computational efficiency, we finally set $d = 64$, as the optimal choice.

In the PLS module, δ has not yet been determined. Therefore, we investigated its impact on the classification results. Treating δ as a similarity threshold, its possible values were restricted to the set {0.65, 0.70, 0.75, 0.80, 0.85, 0.90, 0.95}. The remaining parameters were set as previously described, and L is set to 5. Experiments are conducted sequentially on three publicly available datasets. As shown in Fig. 8, for the IP dataset, the model performs well when the similarity threshold is 0.80, 0.85, and 0.90. For the UP dataset, optimal performance is achieved at thresholds of 0.90 and 0.95. For the LK dataset, the best performance occurs at a threshold of 0.90. The threshold δ is introduced to ensure the reliability

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TABLE III
CLASSIFICATION RESULTS (%) ON IP DATASET

CLASS	TWO-CNN	RN-FSC	SSFTT	DCFSL	Gia-CFSL	CMFSL	DMCM	MambaHSI	CSNPR
C1	66.10	85.43	74.23	94.98	89.27	97.80	**99.51**	99.06	99.27
C2	27.72	31.72	44.87	45.74	49.41	49.84	65.77	**67.75**	67.27
C3	28.18	34.02	40.11	51.72	49.70	58.65	67.05	65.99	**67.33**
C4	31.51	59.41	50.65	75.56	77.11	82.84	88.58	87.30	**91.55**
C5	57.68	50.06	66.73	73.10	74.39	77.16	**81.15**	80.52	**81.15**
C6	58.98	77.79	70.65	86.29	81.77	79.06	92.07	**96.78**	94.97
C7	86.09	96.07	91.27	97.39	99.13	99.56	**100.00**	**100.00**	**100.00**
C8	70.68	88.43	80.37	85.43	90.30	88.67	92.75	91.07	**94.04**
C9	94.00	97.50	98.92	99.33	98.00	**100.00**	**100.00**	**100.00**	**100.00**
C10	32.48	47.92	46.18	59.03	60.46	67.33	70.75	73.83	**74.29**
C11	42.29	38.57	50.66	60.27	61.09	57.42	63.71	71.01	**71.37**
C12	25.41	27.45	38.21	44.52	47.33	46.97	75.29	77.05	**77.69**
C13	91.00	88.34	96.09	97.95	97.45	99.20	99.10	**100.00**	99.20
C14	79.57	76.89	77.43	81.22	79.81	83.58	87.97	87.84	**90.89**
C15	41.23	61.76	51.33	72.31	72.18	75.70	84.96	85.99	**87.14**
C16	79.09	98.39	97.01	97.84	99.43	**99.89**	99.20	98.84	99.20
OA	46.31±3.40	51.26±4.48	53.71±4.82	64.89±3.26	65.21±3.09	66.74±3.31	75.45±4.22	76.31±3.44	**78.79±2.86**
AA	57.00±1.96	66.24±1.88	67.16±2.81	76.41±1.90	77.06±1.92	78.98±1.51	85.49±1.92	86.44±2.18	**87.21±1.48**
Kappa	37.79±3.22	45.75±4.46	48.37±5.63	60.44±3.53	60.88±3.28	62.63±3.36	72.37±4.59	74.33±4.13	**76.02±3.14**

of pseudo-labels during self-learning. A relatively high value ($\delta = 0.9$) is adopted as a conservative lower bound to reduce noise accumulation in early iterations. Such a setting is commonly used in self-training frameworks to balance pseudo-label quantity and quality. Moreover, as demonstrated in the sensitivity analysis, the performance remains stable when δ varies within a reasonable range, indicating that the proposed PLS module does not rely on precise threshold tuning.

C. Comparison of Different Classification Methods

To evaluate the performance of the proposed CSNPR method, several few-shot HSIC methods are selected for comparison, including Two-CNN [37], RN-FSC [35], SSFTT [36], Gia-CFSL, DCFSL, CMFSL, DMCM, and MambaHSI [40]. All these methods belong to the FSL paradigm, among which DCFSL, Gia-CFSL, CMFSL, DMCM, and MambaHSI combine FSL with data analysis techniques. Specifically, SSFTT adopts a transformer-based network for few-shot classification, while MambaHSI employs a Mamba-based network for few-shot classification of HSIs. For all FSL methods, 1–5 samples per class are randomly selected from the TD as labeled samples. Notably, Two-CNN is a semi-supervised method that learns spatial–spectral features through a dual-branch architecture and performs classification fine-tuning on the TD using a source-domain model.

Tables III–V show the classification results of each method on IP, UP, and LK datasets, and the numbers with the highest classification accuracy are thickened. Meanwhile, we show the classification results of each method on IP, UP, and LK datasets in Figs. 9–11, which can view the classification results more intuitively.

1) On the IP dataset, three indicators of the proposed method are 78.79%, 87.21%, and 76.02%, respectively. On the UP dataset, three indicators of the proposed method are 85.36%, 89.12%, and 81.28%, respectively. On the LK dataset, three indicators of CSNPR method are 97.26%, 95.56%, and 96.42%, respectively. On the three public datasets, three indicators of the CSNPR method have achieved the highest values. Among them, the OA value is 2.48% (IP), 1.12% (UP), and 0.83% (LK) higher than the second-ranked method, respectively. There is a significant improvement in the IP and UP datasets, while the LK dataset is not very obvious due to the high OA value of the comparison methods. Clearly, the CSNPR method outperforms the selected methods.
2) Tables III–V have shown that in the case of few-shot samples, the performance of the deep learning method TWO-CNN is worse than that of other FSL models. Even the worst-performing method among the selected FSL methods, its OA value is 4.95% (IP), 1.54% (UP), and 8.12% (LK), higher than that of TWO-CNN. In addition, the FSL method still outperforms the deep learning method in this case.

After analyzing the overall performance, we analyze the experimental results from the perspective of the class. To facilitate the analysis, the number with the highest accuracy of each type is thickened in Tables III–V, and an underscore is added to the second-highest number. On the IP dataset, the CSNPR method makes the classification accuracy of 11 classes of data achieve the highest value, five classes reach the second highest value. On the UP dataset, CSNPR makes the classification accuracy of five classes of data reach the highest value, two classes reach the second highest value, and the remaining two classes of classification accuracy are less than 3% from the highest value. On the LK dataset, CSNPR makes the classification accuracy of six classes of data reach the highest value, two classes reach the second highest value, and the OA value of the remaining one class is 99.5%. Simultaneously, compared with the best performance method, the AA values of the CSNPR method increased by 1.72% (IP), 2.06% (UP), and 1.61% (LK), respectively. In the IP dataset, the C2 class samples are highly similar to the adjacent categories in terms of spectral features, and there is also a problem of unbalanced sample quantity, making them

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TABLE IV
CLASSIFICATION RESULTS (%) ON UP DATASET

CLASS	TWO-CNN	RN-FSC	SSFTT	DCFSL	Gia-CFSL	CMFSL	DMCM	MambaHSI	CSNPR
C1	51.83	58.81	60.41	76.37	79.61	77.41	85.82	88.06	**88.92**
C2	81.93	68.78	76.52	84.63	**84.95**	82.62	81.33	83.70	82.23
C3	36.54	66.61	58.19	63.35	55.69	62.04	72.68	**80.92**	78.31
C4	87.29	88.22	89.88	92.78	93.21	91.84	93.67	90.06	**94.68**
C5	91.52	98.22	97.22	98.75	97.81	99.66	99.34	**99.89**	99.70
C6	38.58	64.96	49.59	77.82	66.48	77.31	77.62	70.73	**78.71**
C7	65.89	67.04	68.56	75.69	73.55	83.33	89.21	90.78	**93.04**
C8	35.72	51.46	50.37	62.15	62.61	76.24	85.65	84.18	**89.02**
C9	96.73	94.67	**99.17**	99.01	98.94	98.64	98.25	98.81	97.49
OA	66.48±4.37	68.02±7.87	67.42±5.66	80.64±4.73	79.54±2.60	81.20±3.55	83.60±4.92	84.24±4.67	**85.36±5.62**
AA	65.12±1.93	73.20±1.92	68.65±1.75	81.18±1.83	79.20±1.60	83.23±1.17	87.06±2.08	87.46±3.62	**90.12±2.58**
Kappa	56.62±4.43	60.17±8.51	57.25±4.97	75.06±5.62	73.43±3.12	75.81±4.11	78.99±5.55	79.31±4.36	**81.28±6.45**

TABLE V
CLASSIFICATION RESULTS (%) ON LK DATASET

CLASS	TWO-CNN	RN-FSC	SSFTT	DCFSL	Gia-CFSL	CMFSL	DMCM	MambaHSI	CSNPR
C1	85.56	97.49	87.42	99.26	98.09	98.06	99.25	**99.63**	99.29
C2	78.31	81.38	80.99	93.51	91.22	91.09	96.88	**97.81**	**97.12**
C3	63.27	85.29	79.47	95.50	93.32	94.63	97.44	95.22	**98.42**
C4	63.43	77.51	74.31	92.55	89.97	89.81	93.76	76.28	**95.45**
C5	89.94	88.23	91.55	90.44	90.21	86.89	94.81	89.25	**95.06**
C6	82.94	92.77	86.84	95.31	92.14	89.92	**97.03**	98.59	96.29
C7	**99.87**	99.80	99.83	99.86	99.82	99.77	99.74	98.91	99.51
C8	58.52	74.35	66.11	81.86	74.50	85.07	85.43	89.25	**93.58**
C9	62.39	71.70	72.43	77.82	76.15	78.83	81.21	93.20	**85.33**
OA	81.19±1.76	89.31±3.58	85.89±4.18	95.54±0.72	93.91±1.05	94.10±0.97	96.43±1.44	94.94±1.46	**97.26±1.29**
AA	76.02±2.57	85.39±3.45	82.10±3.77	91.79±2.38	89.49±2.48	90.45±2.97	93.95±2.50	93.13±2.42	**95.56±1.20**
Kappa	76.22±2.15	89.32±4.41	84.23±1.67	94.17±0.92	92.07±1.34	92.31±1.25	95.35±1.86	93.75±3.16	**96.42±1.67**

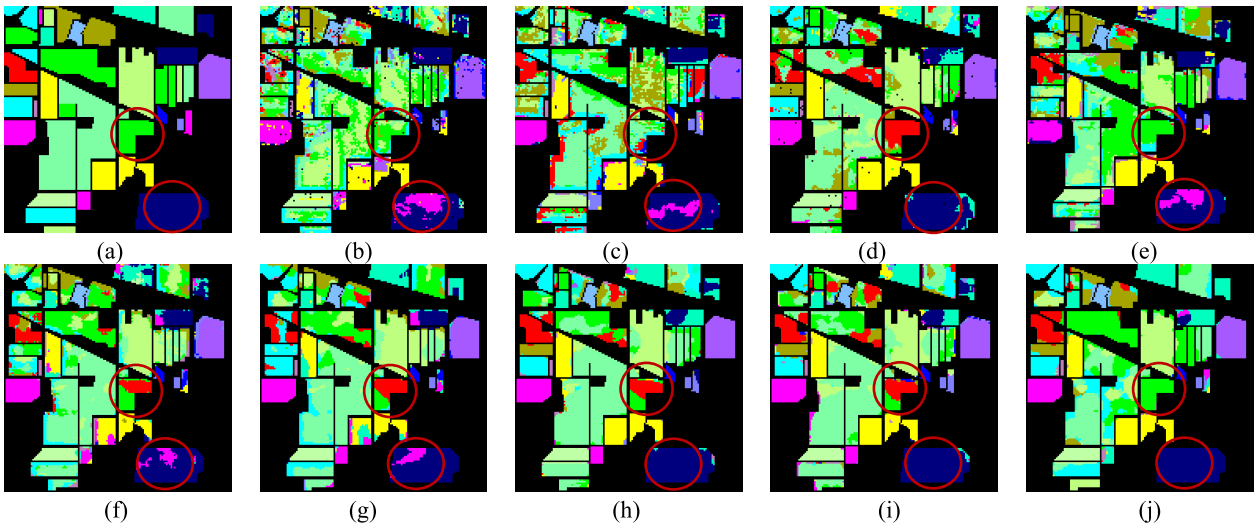


Fig. 9. Classification visualization maps of different models on the IP dataset. (a) Ground. (b) Two-CNN. (c) RN-FSC. (d) SSFTT. (e) DCFSL. (f) Gia-CFSL. (g) CMFSL. (h) DMCM. (i) MambaHSI. (j) CSNPR.

particularly sensitive to class prototypes shift and domain deviation. CSNPR significantly enhances the discrimination ability of this class through prototype correction and a class-aware domain alignment mechanism, thus showing a more significant improvement on the C2 class. In contrast, the LK dataset has higher spatial resolution and clearer class boundaries. Most comparison methods have achieved near saturation in classification performance, so the improvement of CSNPR on some LK classes is relatively limited, but it still maintains a stable advantage in OA and AA. In Figs. 9–11, we circle some areas in the figure with red circles to analyze the results. Figs. 9 and 10 show that the red circle area has fewer and fewer miscellaneous colors, and the colors are gradually unified, indicating that there are fewer misclassifications in the same category and better performance. As for Fig. 11, it represents the classification results of each method on the LK

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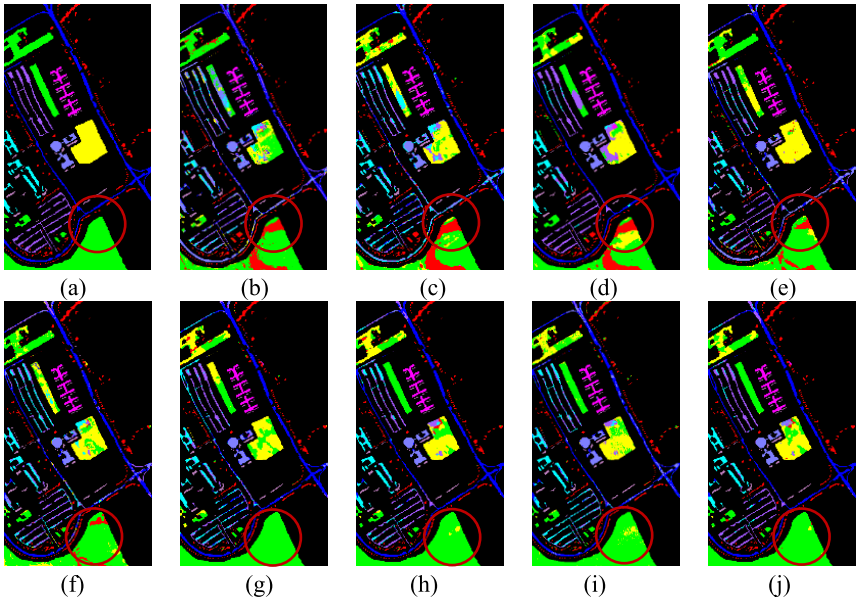


Fig. 10. Classification visualization maps of different models on the UP dataset. (a) Ground. (b) Two-CNN. (c) RN-FSC. (d) SSFTT. (e) DCFSL. (f) Gia-CFSL. (g) CMFSL. (h) DMCM. (i) MambaHSI. (j) CSNPR.

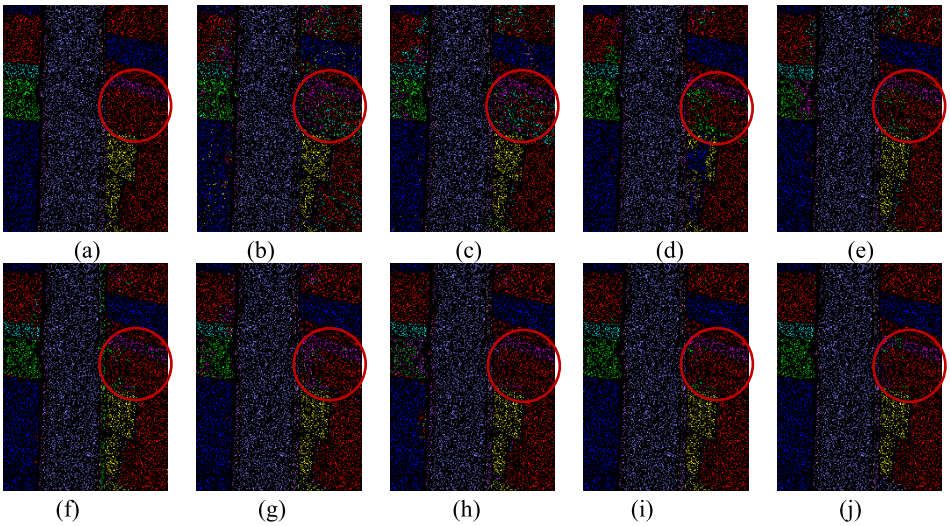


Fig. 11. Classification visualization maps of different models on the LK dataset. (a) Ground. (b) Two-CNN. (c) RN-FSC. (d) SSFTT. (e) DCFSL. (f) Gia-CFSL. (g) CMFSL. (h) DMCM. (i) MambaHSI. (j) CSNPR.

dataset. The OA of the LK dataset is high, so there is no obvious discrepancy.

In summary, the classification performance of the CSNPR method is superior to other selected methods, both in terms of overall indicators and class indicators.

D. Ablation Experiment

To evaluate the effectiveness of each module, ablation experiments were conducted on three publicly available datasets: IP, UP, and LK datasets. The results of these experiments are presented in Table VI. Among them, A represents the ACDAA strategy, B denotes the TGAN-B network employing the prototype rectification-based FSL strategy, and S is the PLS module. The models used in this section are DMCM with TGAN network and DCFSL with traditional CDAA strategy. As can be seen from the table, the OA of IP, UP, and LK of the DCMM method increased by 0.4%, 0.68%, and 0.39%, respectively, after the ACDAA strategy was employed.

Considering that the DMCM method does not adopt the CDAA strategy, we conducted a comparative experiment using the DCFSL method. The OA of IP, UP, and LK increased by 1.26%, 1.99%, and 0.46%, respectively. This indicates that the ACDAA strategy has taken effect. After employing the prototype rectification-based TGAN-B network, the OA increased by 1.0%, 0.64%, and 0.09% on the IP, UP, and LK datasets, respectively. Although the proposed method achieves obvious performance improvements on the IP and UP datasets, the performance gain on the LK dataset is relatively limited. This is mainly because the core contribution of CSNPR lies in proposing more efficient mechanisms for feature modeling and domain alignment, rather than merely pursuing minor

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TABLE VI
ABLATION STUDY (%) ON THE OA AND AA INDEX FOR DATASETS

		DMCM	DMCM+CDAA	DMCM+A	DMCM+B	DMCM+S	DMCM+B+S	CSNPR(CDAA)	CSNPR
IP	OA	75.45	75.63	75.85	76.45	76.72	77.65	77.82	**78.79**
	AA	85.49	85.72	85.84	86.09	86.41	86.75	87.03	**87.21**
UP	OA	83.60	84.10	84.28	84.24	84.32	84.77	84.96	**85.36**
	AA	87.06	87.56	88.28	89.46	89.77	89.92	89.43	**90.12**
LK	OA	96.43	96.62	96.82	96.52	96.59	96.72	96.99	**97.26**
	AA	93.95	94.07	94.92	94.45	94.53	94.64	94.87	**95.56**

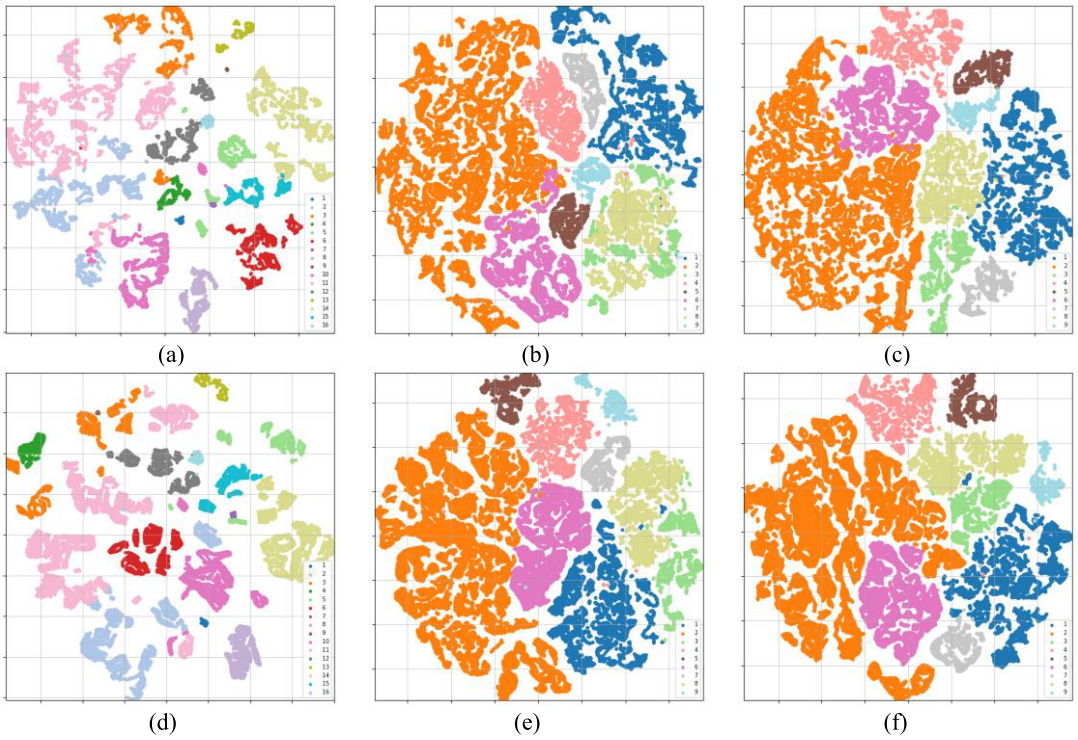


Fig. 12. Feature visualization of three dataset with the t-SNE. (a) DMCM (IP). (b) DMCM (UP). (c) DMCML (LK). (d) DMCM + B (IP). (e) DMCM + B (UP). (f) DMCM + B (LK).

TABLE VII
TIME METRICS OF THE METHODS

		DCFSL	DCFSL+A	DMCM	DMCM+B	DMCM+B+S	CSNPR
IP	Train time(s)	580.43	632.29	2516.43	2343.67	2744.88	3067.49
	Test time(s)	5.03	5.12	0.83	0.90	0.85	1.02
UP	Train time(s)	558.19	606.73	1957.69	1742.78	2196.59	2537.92
	Test time(s)	1.26	1.19	3.72	3.74	3.68	3.99
LK	Train time(s)	741.58	793.40	3106.71	2855.91	3287.38	3508.14
	Test time(s)	6.47	6.52	1.13	1.08	1.10	1.21

accuracy improvements on performance-saturated datasets. Accordingly, the improvements of each module are restricted to the LK dataset with low classification difficulty, while the proposed method achieves significant and stable performance gains on complex datasets with fewer samples and more pronounced domain discrepancies.

Fig. 12 illustrates the changes in feature visualization before and after applying TGAN-B: on the IP dataset, features from the same class become more tightly clustered with clearer boundaries; a similar effect is observed on the UP dataset; On the LK dataset, where class boundaries were already distinct, only a slight tightening of intra-class features is evident. These findings confirm that the module effectively calibrates class prototypes. Considering that the semi-supervised strategy benefits more from a thorough extraction of SD knowledge, we did not conduct experiments by adding the PLS module separately. Instead, the PLS module was incorporated on top of the TGAN-B network. The function of the PLS module is to filter out the pseudo-label samples with high confidence by using the intra-class similarity of the TD samples. Its effectiveness depends on the premise that "the feature extractor has been able to learn discriminative features." If the PLS

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TABLE VIII
IMPACT OF NOISY ANOMALIES IN TD ON THE BDC MODULE

		DMCM			DMCM+B		
IP	unseen classes in SD	Class 8	Class 15	Class 16	Class 8	Class 15	Class 16
	OA	92.75	84.96	99.20	93.17	86.03	99.43
UP	unseen classes in SD	Class 2	Class 5	Class 7	Class 2	Class 5	Class 7
	OA	81.33	99.34	89.21	81.69	99.52	91.25
LK	unseen classes in SD	Class 3	Class 4	Class 5	Class 3	Class 4	Class 5
	OA	97.44	93.76	94.81	97.89	94.41	94.93

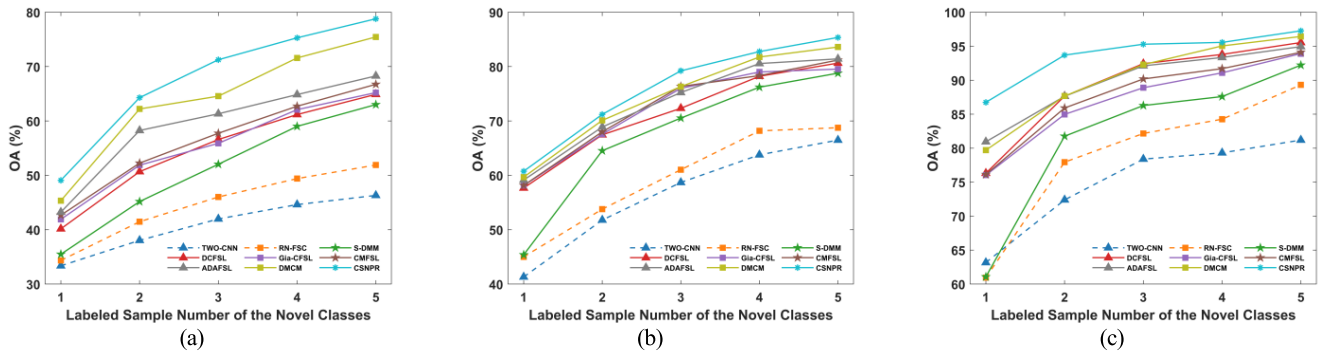


Fig. 13. Classification accuracy (OA) with different numbers of labeled samples on target datasets. (a) IP. (b) UP. (c) LK.

module is tested alone, due to the insufficient discriminative power of the original features, the initial allocation accuracy of the pseudo-labels is low. At this time, even if the PLS module filters out "relatively high confidence" samples, it may contain a large number of incorrect labels, and thus cannot objectively reflect the true performance of this module. The TGAN-B network optimizes the feature expression through GA and the BDC module, significantly improving the reliability of the initial pseudo-labels. Only by adding the PLS module can its "purification of training samples" function be accurately verified. Therefore, to ensure the rationality of the experiment and the objectivity of the results, we choose to test the gain of the PLS module on the basis of the TGAN-B network, rather than testing it alone. After adding the PLS module, the OA value further improved by 1.2% (IP), 0.53% (UP), and 0.2% (LK), indicating that this module is also effective. Finally, by integrating all the aforementioned modules into the DMCM method, we developed the CSNPR method, which achieved the highest OA and AA values. This outcome suggests that the modules do not interfere with each other negatively, or any negative interference is minimal.

The time metrics of the methods used for ablation are shown in Table VII. The ACDAA method introduces DA weights on top of CDAA, resulting in a slight increase in computational cost; however, the overall time overhead is not significantly increased. DCFSL + A exhibits a marginal rise in time metrics compared to the baseline DCFSL, but this increase is almost negligible. In addition, the proposed TGAN-B method effectively reduces the computational cost, achieving lower time consumption than the baseline TGAN. Benefiting from the time efficiency of TGAN-B, the overall time metrics of DMCM + B + S are slightly higher than those of DMCM, but the classification accuracy is improved.

The BDC module captures high-order dependencies among spectral features, providing a more discriminative feature representation for the proposed method. To further evaluate its effectiveness in the presence of anomalies in the TD, particularly unseen classes that are not involved in the SD, we analyze its classification performance on the newly emerged classes in three TD datasets. The corresponding results are presented in Table VIII.

The experimental results demonstrate that, after incorporating the BDC module, the classification performance on the PU, IP, and LK datasets is improved to varying degrees, not only for the shared classes between the source and TDs but also for the unknown classes (i.e., anomalies) appearing in the TD. This indicates that the BDC module can effectively model complex statistical dependencies among features, enabling the model to learn discriminative representations even in the presence of unknown class interference and noise, thereby exhibiting strong robustness.

Overall, these three components significantly enhance the performance of classification under scenarios with sparse labeled samples, although their contributions vary across different datasets due to their respective focuses.

E. Influence of the Number of Labeled Samples in the TD

The quantity of labeled samples from the TD is critical in determining classification accuracy. Generally, a larger number of labeled samples provides more informative supervision, leading to improved performance. In this section, L labeled samples per class are chosen from the TD, with $L \in \{1, 2, 3, 4, 5\}$, to simulate different levels of supervision. The impact of L on classification accuracy is evaluated based on the OA obtained by various methods. Fig. 13 illustrates the

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OA values of different methods on the IP, UP, and LK datasets as the number of labeled samples in the TD increases.

Apparently, as the L value increases, OA values of all classification models will also increase, which is the same as expected. Another thing that can be seen is that the FSL method has a smooth change in OA when the L is 2–5, but the number will change greatly when it changes from 2 to 1, indicating that the FSL method also needs the support of a certain number of labeled samples. Finally, when L is 1–5, CSNPR has higher classification accuracy than other selected methods.

IV. Conclusion

In this article, we propose a cross-domain prototype-rectified framework, termed CSNPR, for few-shot HSIC, with the goal of addressing two fundamental challenges in this task: inaccurate class prototypes and class-agnostic domain alignment under severe sample scarcity. To this end, TGAN-B is designed to efficiently learn discriminative spectral–spatial representations of HSIs while maintaining low computational cost, and to refine class prototypes through an enhanced metric-based FSL strategy. To further alleviate domain discrepancies at the class level, the ACDAA strategy is introduced, in which class-wise adaptive weights are assigned according to the degree of distributional discrepancy between the source and TDs, enabling the model to focus on classes with larger domain shifts. In addition, the PLS module is incorporated to selectively expand the labeled training set in the TD by exploiting high-confidence predictions and intra-class similarity, thereby enhancing the effectiveness of semi-supervised learning under limited supervision. Extensive experiments conducted on three publicly available hyperspectral datasets demonstrate that, with only scarce labeled samples, CSNPR consistently outperforms state-of-the-art (SOTA) and classical methods in terms of OA, AA, and Kappa coefficient. These results confirm that the proposed framework effectively improves class prototype reliability and achieves more precise class-wise domain alignment, ultimately leading to robust cross-domain few-shot classification performance.

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